

ASSESSMENT-1

1.Question:

Which development methodology would you recommend for this project, and why? Identify two specific risks the team would face if they chose the wrong methodology for this context.

Answer:

I would recommend Agile methodology for this project.

The reason is that the requirements of the digital wallet app can change frequently. Also, the client wants to check working features every two weeks. Agile allows the team to develop the project in small parts and get regular feedback from the client.

Two risks of choosing the wrong methodology:

1. Difficulty in handling changes: If the team chooses Waterfall, changing requirements later can be difficult and may require extra time and work.
2. Delayed feedback: The client may not get to review the working features regularly. If a problem is found later, the team may have to redo a lot of work.

So, Agile is suitable because it supports changing requirements and regular client feedback.

5.Question:

Explain why an array is the correct data structure for this task compared to using 30 separate variables. Describe the two-step logic your program would follow: first to compute the average, then to identify the above-average days — and explain why both steps cannot be done in a single loop.

Answer:

- An array is useful because we can store 30 days' minutes in one array instead of creating 30 separate variables. It makes the program simple and easy to manage.
- First, the program calculates the total minutes and then finds the average. Second, it checks each day's minutes with the average and finds the days having more minutes than the average.
- We cannot normally do both steps in one loop because we need the average first. After getting the average, we can compare each day's minutes with it.

6.Question:

Explain why the program crashes on an empty string input and describe the check your pointer-based solution must perform before dereferencing the pointer. How does traversing a string using a pointer differ from traversing it using array index notation, and which approach makes the empty-input bug easier to catch?

Answer:

- The program can crash when the user gives an empty string because there is no valid character to access.
- If the program dereferences an invalid pointer, it may access invalid memory and crash.
- Before dereferencing a pointer, the program should check that the pointer is valid and the string is not empty.
- With array notation, we use `str[i]` to access characters. With a pointer, we use the pointer and move it through the string.
- Array index notation is easier for beginners to understand and makes the empty-input problem easier to catch.